# APPROXIMATION BY PERFECT COMPLEXES HOLDS ON QCQS ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: the counted statement is the author theorem environment labelled theorem-approximation; the proof environment immediately following it is its own proof

*Editorial note.* The label theorem-approximation names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own theorem environment in the chapter source.

*Allowed standard prerequisites (editorial).*

- the induction principle for qcqs algebraic spaces (lemma-induction-principle, delivered as its own card)
- the induction step lemma of the same chapter (lemma-induction-step) and the affine case lemma (lemma-approximation-affine), both proved in the same chapter

*Source reference.* Original reference 08EQ: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 3651–3651. Prerequisite pointer invoked at 3795; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 08DR: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1563–1563. Prerequisite pointer invoked at 3794; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 05I3: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 3685–3685. Prerequisite pointer invoked at 3795; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Theorem 0.1.** *Let $S$ be a scheme. Let $X$ be a quasi-compact and quasi-separated algebraic space over $S$. Then approximation by perfect complexes holds on $X$.*

*Proof.* This follows from the induction principle of Lemma 08DR and Lemmas 05I3 and 08EQ. □

*Date*: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.